PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)
(Established under the Panjab University Act VII of 1947- enacted by the Government of India)

FACULTY OF ARTS

SYLLABI

FOR

M.A. GEOGRAPHY

(Two Year Degree Programme, 1st & 2nd Semester System)

NEP 2020
Department of Geography, Panjab University Chandigarh
Implemented from the Academic Session 2026-27

Examinations, 2026-27

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M.A. GEOGRAPHY
(Two Year Degree Programme, Semester System)
PROGRAM OUTCOMES

M.A. Geographyis atwo-year postgraduate programme that amalgamates science and humanities to lay the foundations of geography as a truly multidisciplinary body of knowledge. M.A. Geography at Panjab University is a popular postgraduate degree programme that offers philosophical, conceptual, theoretical, methodological and applied knowledge on physical and human facets of geography. The course encompasses innovatively designed rigorous coursework that underscores research training and fieldwork-based knowledge augmentation. It covers subject matter in a comprehensive and scientific manner. The course content and learning outcomes offers innovative teaching-learning experience aiming at academic, research, and professional growth of students to meet the goals of Indian education system for inclusive development of the country.

Keeping pace with changing global, national and local scenario and recognising the need to transform our society's sustainability, the program offers innovative, multi-disciplinary, flexible and robust course content that caters different dimensions of geographic knowledge. This two-year programme divided into four semesters are designed to inculcate intellectual, philosophical,professional attitude and skills. Students are exposed to a wide range of knowledge, skill enhancement and character-building practices. We offer a spectrum of learning and skills ranging from conceptual and theoretical clarity, technical and analytical capabilities, grasp on physical, social, economic, political and environmental contexts of geographic knowledge. Trained through a focused field-work based learning and research orientation, we prepare students to synthesize knowledge and practical understanding of processes, mechanisms and operational aspects of Geography.

Program Learning Outcomes: Students will develop the ability to:

- Comprehend philosophy, key concepts and methodologies in geography.
- Analyse mechanisms and processes associated with physical and human aspects of geography.
- Use the tools of cartography, RS, GIS, GNSS, and Surveying for mapping and analysis.
- Demonstrate spatial analysis and research skills in the form of tangible products.
- Apply technical skills and field-knowledge to problem solving.
- Create outputs to meet local, regional, and global goals of development.
- Assess and critique local, regional and global scenarios, practices and policies.

PROGRAMME OUTLINE
M.A. GEOGRAPHY (TWO YEAR PROGRAMME)

Course Type	Course Code	Course Title	Max. Marks	Credit
Semester I				
Discipline Specific Core (DSC)	GEOG 101	Evolution of Geographic Thought	100	4
	GEOG 102	Geomorphology	50	2
	GEOG 103	Geomorphological Techniques (Practical)	50	2
	GEOG 104	Cartography	50	2
	GEOG 105	Cartographic Techniques and Visualisation (Practical)	50	2
Discipline Specific Elective (DSE)	GEOG 106	Any one of the Following: i. Geography and Ecosystems ii. Population Geography	100	4
	GEOG 107	Any one of the Following: i. Cultural Geography ii. Electoral Geography	100	4
		Semester Total	***500***	***20***
Semester II				
Discipline Specific Core (DSC)	GEOG 201	Geography of India	100	4
	GEOG 202	Climatology	50	2
	GEOG 203	Climate and Weather Analysis (Practical)	50	2
	GEOG 204	Fundamentals of Remote Sensing	50	2
	GEOG 205	Digital Image Processing (Practical)	50	2
Discipline Specific Elective (DSE)	GEOG 206	Any one of the Following: i. Agriculture Geography ii. Hazards and Disasters	100	4
	GEOG 207	Any one of the Following: i. Administrative Geography ii. Urban Geography	100	4
		Semester Total	***500***	***20***
Semester III				
Discipline Specific Core (DSC)	GEOG 301	Regional Planning	100	4
	GEOG 302	Fundamentals of GIS and GNSS	50	2
	GEOG 303	GIS Tools and Techniques (Practical)	50	2
RBC-RM	GEOG 304	Research Methodology in Geography	50	2
	GEOG 305	Research Tools and Techniques (Practical)	50	2
Discipline Specific Elective (DSE)	GEOG 306	Any one of the Following i. Biogeography ii. Population Development and Planning iii. Regional Development and Planning in India iv. Special Themes in Agricultural Geography	100	4
Discipline Specific Core (DSC)	GEOG 307	GNSS and Surveying (Practical)	50	2
	GEOG 308	Geographic Thought- A Critical Introduction	50	2
		Semester Total	***500***	***20***
Semester IV				
Discipline Specific Core (DSC)	GEOG 401	Quantitative Methods	50	2
	GEOG 402	Spatial Analysis in Geography (Practical)	50	2
RBC-RM	GEOG 403	Qualitative Research Methods	100	4
Discipline Specific Elective (DSE)	GEOG 404	Any one of the Following i. Climate Change and Earth System ii. Geography of Water Resources iii. Migration Geography iv. Social Geography	100	4
Discipline Specific Core (DSC)	GEOG 405	Dissertation (Fieldwork Based Research)	100	4
	GEOG 406	Contemporary Human Geography	100	4
		Semester Total	***500***	***20***
		GRAND TOTAL (Semester I, II, III, IV)	***2000***	***80***

Semester-I

DISCIPLINE SPECIFIC CORE (DSC) COURSES

GEOG 101
EVOLUTION OF GEOGRAPHIC THOUGHT

Course Code: GEOG 101
Course Type: DSC
Course Credit: 4
Teaching Hours/Week: 4

Evaluation: Maximum Marks: 100
Terminal Examination: 80 Marks
Internal Assessment: 20 Marks
Examination Time: 3 Hours

COURSE OUT COMES: At the end of the course the students will be able to:

- Synthesize the history, philosophy and scope of Geography
- Recount the development of Geographic thinking and link it to the development of knowledge
- Assess and critique the development of geographical paradigms, debates and approaches
- Demonstrate the skills of critical thinking, lucid writing and effective presentation.

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 short questions in about 25-30 words each. Each question shall carry 2 marks (Total 20 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 15 marks (Total 60 marks). These will be in addition to the compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. **The paper setter must put note (4) in the question paper.**

COURSE CONTENT

Unit-I

i. ContributionsoftheGreeks&Romansw.r.t. Herodotus,Eratosthenes, StraboandPtolemy
ii. Geography intheMiddleAges:ContributionsofArabs;DarkAgesinEurope
iii. GeographyandtheRenaissance;Geographyandtheageofexplorationsanddiscoveries

Unit-II

iv. Pre-ClassicalandClassicalGeography:Varenius,Kant,vonHumboldtandRitter
v. DarwinisminGeography,EnvironmentalDeterminismandPossibilism
vi. RegionalGeography: FrenchandBritishSchool, GeographyasstudyofArealDifferentiation

Unit-III

vii. Dichotomies in Geography
   - Regional vs. Systematic Geography
   - Physical vs. Human Geography
   - Idiographic vs. Nomothetic Approach

Unit-IV

viii. QuestforaScientificFlavour
   - Positivism,SchaeferandGeographyasaSpatial Science,QuantitativeRevolution
   - ScientificMethodinGeography
   - CriticismofPositivism,Radicalism,Humanism,Behavioralism

LISTOFREADINGS

EssentialReadings:

1. Gosal,G.S.:*HistoryofGeographicThought*,PanjabUniversityPress, Chandigarh,2015.
2. Nayak,A.andA.Jeffrey.:*GeographicalThought*,Routledge,LondonandNewYork,2016.

3. Dikshit,R.D.(ed.):*GeographicalThought:AContextualHistoryofIdeas*,PrenticeHallofIndia, NewDelhi, 1999.
4. Hartshorne,R.: *PerspectivesontheNatureofGeography*,RandMcNally&Co.,1959.
5. Harvey,D.:*ExplanationinGeography*,EdwardArnold,London,1973.
6. Martin,G.:*AllPossibleWorlds,AHistoryofGeographicalIdeas*,4th Edition,OxfordUniversityPress, NewYork, 2005.
7. Peet,Richard:*ModernGeographicalThought*,Blackwell,Massachusetts,FirstIndianReprint,2004.
8. Jensen,A.H.:*Geography:ItsHistoryandConcepts*,SagePublications,ThousandOaks,New Delhi,2009.
9. Livingstone,D.*TheGeographicalTradition*,Blackwell,Oxford,1993.

Suggested Readings:

1. Abler,RonaldF.Marcus,Melvin,G.Olson,Judy,M.:*Geography'sInnerWorldsPervasiveThemesinContemporaryAmericanGeography*,RutgersUniversityPress,NewJersey,1992.
2. Abler,Ronald,Adams,John S.and Gould,Peter:*SpatialOrganization:TheGeographer'sViewoftheWorld*,N. J., PrenticeHall,1971.
3. Board,Christopher,R.J.,Haggett,P.,Stoddart,D.R.(ed.):*ProgressinGeography:InternationalReviewofCurrentResearch*,Vol.ItoVIII,EdwardArnold,London.
4. Johnston,R.J.&Claval,P.:*GeographySincetheSecondWorldWar:AnInternationalSurvey*,CrownHalm,Sydney, 1984.
5. Johnston,R.J.andSidaway,J.D.:*GeographyandGeographers: Anglo-AmericanHumanGeographySince1945*,Arnold,London,2004.
6. Johnston,R.J.:*PhilosophyandHumanGeography*,EdwardArnold,London,1983.
7. Minshull,R.:*The ChangingNature ofGeography*,Hutchinson,University Library,London,1970.

GEOG 102
GEOMORPHOLOGY

Course Code: GEOG 102
Course Type: DSC
Course Credit: 2
Teaching Hours/Week: 2

Evaluation: Maximum Marks: 50
Terminal Examination: 40 Marks
Internal Assessment: 10 Marks
Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Describe concepts, theories and models of geomorphological knowledge
- Explain the interactions between geomorphic processes and landform evolution
- Comprehend the interactions between endogenic and exogenic processes
- Identify and interpret landforms

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 6 short questions in about 25-30 words each. Each question shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to the compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. **The paper setter must put note (4) in the question paper.**

COURSE CONTENT

Unit-I

i. Nature, scope and approaches of Geomorphology.
ii. Concepts of time, scale, and threshold in landform studies.
iii. Earth's Interior and Isostasy (Airy and Pratt).

Unit–II

iv. Fundamental Principles: Uniformitarianism, Geomorphic Cycles (Davis's and Penck's models), and Geomorphic Systems (equilibrium, process-response).
v. Geomagnetism: Characteristics, importance and applications of geomagnetism.
vi. Denudation chronology: Channel morphology, erosion surfaces, and slope development.

Unit-III

vii. Applied geomorphology: Understanding landslides, floods, coastal erosion, and desertification.
viii. Geomorphology in soil conservation, cryosphere dynamics, and mineral exploration.
ix. Terrain analysis for infrastructure planning (dams, roads) and environmental impact assessment.

Unit-IV

x. Field Methods: Basic surveying and terrain data collection.
xi. Geospatial Techniques: Application of GIS (Geographic Information Systems), Remote Sensing (RS), and GPS in mapping, monitoring, and modeling landform changes.
xii. Use of Geospatial AI (ML and DL) in geomorphology.

LIST OF READINGS

Essential Readings:

1. Bierman, P. and Montgomery, D.: *Key Concepts in Geomorphology - 14 edition,* W.H. Freeman, U.K. 2014.
2. Bloom, Arthur L.: *Geomorphology: A Systematic Analysis of Late Cainozoic Landforms*, Pearson Education, Singapore, 3rd Edition, 2003.
3. Bridge, J.S.: *Rivers and Floodplains: Forms, Processes, and Sedimentary Record*, Wiley-Blackwell, UK, 2003.
4. Bridge, J. and Demicco, R.: *Earth Surface Processes, Landforms and Sediment Deposits,* Cambridge University Press, New York, 2008.
5. Goudie, A. and Viles, H.: *Landscapes and Geomorphology: A Very Short Introduction,* Oxford University Press, 2010.
6. Gregory, K.J.: *The Earth's Land Surface: Landforms and Processes in Geomorphology*, Sage Publications (CA), 2010.
7. Leopold, L.B., Miller, J.P. and Wolman M.G.: *Fluvial Processes in Geomorphology,* Dover Publications, 1995.
8. Richard, J.H.: *Fundamentals of Geomorphology*, Routledge, Taylor & Francis Group, London, 2011.
9. Singh, Savindra: *Geomorphology*, Prayag, Prakashan, Allahabad, 1998.

Suggested Readings:

1. Chorley, R.J., Beckinsale, R.P. and Dunn, A.J.: *The History of the Study of Landforms or The Development of Geomorphology*, Volume Two: The Life and Work of William Morris Davis,Methuen & Co. Ltd., London, 1973.
2. Chorley, R.J. and Beckinsale, R.P.: *The History of the Study of Landforms or The Development of Geomorphology,* Volume 3: Historical and Regional Geomorphology 1890–1950, Routledge,London, 1991.
3. Davis, W. M.: *Geographical Essays*, Dover, Boston, 1909.
4. King, L.C.: *The Morphology of the Earth*, Hafner, New York, 1962.
5. Leopold, L.B.: *Water, Rivers and Creeks,* University Science Books, 1997.
6. Leopold, L.B.: *A View of the River,* Harvard University Press, 2006.
7. Penck, W.: *Morphologic Analysis of Landforms*, St. Marisip Press, London, 1953.
8. Strahler, Alan and Arthur Strahler: *Physical Geography: Science and Systems of the Human Environment*, John Wiley & Sons, New York, 3rd Edition, 2005.
9. Thornbury, W. D.: *Principles of Geomorphology*, John Wiley, New York, 1969.
10. Twidale, C.R.: *Structural Landforms*, A.N.U. Press, Canberra, 1971.

GEOG 103 (Practical)
GEOMORPHOLOGICAL TECHNIQUES

Course Code: GEOG 103 (Practical)
Course Type: DSC
Course Credit: 2
Teaching Hours/Week: 4

Evaluation: Maximum Marks: 50
Practical Record: 30 Marks
Viva-Voce: 10 Marks
Internal Assessment: 10 Marks
Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Collect geomorphic information from topographic maps, satellite imageries and field studies
- Analyse and interpret data using various tool and techniques
- Delineate distinct landscape units and features
- Evaluate topographical characteristics and underlying geomorphic processes

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: Practical Exercises/Lab-Work

1. Construction and interpretation of Isometric Block Diagram of landforms
2. Mapping bedding plane, dip, strike, folds, faults and associated features
3. Using maps, satellite data, google earth for landform identification and demarcation
4. Study of fluvial landforms and processes using satellite imageries and field work
5. Measuring aspects of drainage basin and stream morphometry
6. Case Study: Mapping topographic and geomorphic attributes

LIST OF READINGS

Essential Readings:

1. Bridge, J. and Demicco, R.: *Earth Surface Processes, Landforms and Sediment Deposits,* Cambridge University Press, New York, 2008.
2. Gregory, K.J.: *The Earth's Land Surface: Landforms and Processes in Geomorphology*, Sage Publications (CA), 2010.
3. Goudie, A.: *Geomorphological Techniques,* Second Edition, Routledge, New York, 1990.
4. Kondolf, G.M. and Piegay, H.: *Tools in Fluvial Geomorphology,* Wiley, Chichester, 2003.
5. Monkhouse, F.J. and Wilkinson, H.R.: *Maps and Diagrams: Their Compilation and Construction,* B.I. Publications Pvt. Ltd., New Delhi, 1994
6. Leopold, L.B., Miller, J.P. and Wolman M.G.: *Fluvial Processes in Geomorphology,* Dover Publications, 1995.
7. Richard, J.H.: *Fundamentals of Geomorphology*, Routledge, Taylor & Francis Group, London, 2011.

Suggested Readings:

1. Leopold, L.B.: *Water, Rivers and Creeks,* University Science Books, 1997.
2. Leopold, L.B.: *A View of the River,* Harvard University Press, 2006.
3. Small, R. J.: *The Study of Landforms*, Cambridge University Press, Cambridge, 1970.
4. Twidale, C.R.: *Structural Landforms*, A.N.U. Press, Canberra, 1971.
5. Wang, Chi-Yuen and Manga, M.: *Earthquakes and Water*, Lecture Notes in Earth Sciences,Springer- Verlag Berlin Heidelberg, 2010.

GEOG 104
CARTOGRAPHY

Course Code: GEOG 104 ***Evaluation: Maximum Marks: 50***

Course Type: DSC — Terminal Examination: 40 Marks

Course Credit: 2 — Internal Assessment: 10 Marks

Teaching Hours/Week: 2 — Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Trace and contextualize the evolution of cartography from the ancient times to the present
- Deconstruct and classify theoretical principles of cartography
- Develop skills for map designs using cartographic principles
- Apply cartographic knowledge for representation and interpretation of information.

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 6 short questions in about 25-30 words each. Each question shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to the compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. **The paper setter must put note (4) in the question paper.**

COURSE CONTENT

Unit I

i. Cartography: Nature, scope and historical evolution
ii. Computer Assisted Cartography

Unit II

iii. Types of Data and Data Sources
iv. Elements of Map Reading: Interpreting Physical and Cultural Features

Unit -III

v. Elementary conventional methods: Contours, Spot heights, Hachures, Hill shading
vi. Profiles: Transverse and Longitudinal

Unit -IV

vii. Representation of Population data and Agricultural data
viii. Representation of Climatic data: Temperature and Rainfall

LIST OF READINGS

Essential Readings

1. Crampton, W. C.: Mapping: A Critical Introduction to Cartography and GIS, John Willy &Sons, New York, 2010.
2. Dodge Martin.: Cartographic Principles: Map Design,Routledge, 2015
3. Field, Kenneth, Cartography, Esri Press, 2018
4. Harvey, Francis: A Primer of GIS, Fundamental Geographic and Cartographic Concepts, The Guilford Press, 2008.
5. Maltiar K.K and Maltiar S.R., Concepts of Cartography, Remote Sensing and GIS, Rajesh Publications, 2019
6. Robinson, A.H. and Others: Elements of Cartography, John Wiley & Sons, New York, 6th Edition, 1992.

Suggested Readings

1. Heywood, I., Cornelius, S., and Carver, S.: An Introduction to Geographical Information

Systems, Prentice Hall, 3rd Edition, 2006.
2. John Krygier and Denis Wood: Making Maps: A Visual Guide to Map Design for GIS, Guilford Publications, 2013.
3. Misra, R.P. and Ramesh, A.: Fundamental of Cartography, Concept Publishing Company, New Delhi, 2014
4. Monkhouse, F.J.: Maps and Diagrams, Methuen and Co., London, 1994.
5. Singh, R. L. and Singh, R.P.B.: Elements of Practical Geography, Kalyani Pub., Delhi, 2005.

GEOG 105 (Practical)
CARTOGRAPHIC TECHNIQUES AND VISUALISATION

Course Code: GEOG 105 (Practical) ***Evaluation: Maximum Marks: 50***

Course Type: DSC — Practical Record: 30 Marks
Course Credit: 2 — Viva-Voce: 10 Marks
Teaching Hours/Week: 4 — Internal Assessment: 10 Marks
Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Demonstrate practical understanding of mapping techniques and map interpretation
- Decipher mapped data and demonstrate cognitive skills
- Develop graphic communication and presentation skills
- Design map outputs using cartographic principles.
- Relate map information to the real world in spatial context.

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. A continuous assessment shall take place during the course for a regular evaluation of the learning The internal assessment shall be based on written assignments, snap tests, term papers, participation, knowledge/skills gained, aptitude, group participation, attendance, and timely completion of tasks/assignments.

COURSE CONTENT: Practical Exercises/Lab Work

1. Methods of landform analysis: Contours, Spot heights, Hachures, Hill shading
2. Profiles and cross sections: Transverse and Longitudinal
3. Methods of slope analysis and interpretation(Wentworth and Robinson)
4. Mapping geographical data: Choropleth and Dot Method
5. Constructing statistical diagrams: Bar Graphs, Line Graphs, Pie Charts

LIST OF READINGS

Essential Readings

1. Crampton, W. C.: Mapping: A Critical Introduction to Cartography and GIS, John Willy &Sons, New York, 2010.
2. Dodge Martin.: Cartographic Principles: Map Design,Routledge, 2015
3. Field, Kenneth, Cartography, Esri Press, 2018
4. Harvey, Francis: A Primer of GIS, Fundamental Geographic and Cartographic Concepts, The Guilford Press, 2008.
5. Maltiar K.K and Maltiar S.R., Concepts of Cartography, Remote Sensing and GIS, Rajesh Publications, 2019
6. Robinson, A.H. and Others: Elements of Cartography, John Wiley & Sons, New York, 6th Edition, 1992.

Suggested Readings

1. Heywood, I., Cornelius, S., and Carver, S.: An Introduction to Geographical Information Systems, Prentice Hall, 3rd Edition, 2006.
2. John Krygier and Denis Wood: Making Maps: A Visual Guide to Map Design for GIS, Guilford Publications, 2013.
3. Misra, R.P. and Ramesh, A.: Fundamental of Cartography, Concept Publishing Company, New Delhi, 2014
4. Monkhouse, F.J.: Maps and Diagrams, Methuen and Co., London, 1994.
5. Singh, R. L. and Singh, R.P.B.: Elements of Practical Geography, Kalyani Pub., Delhi, 2005.

DISCIPLINE SPECIFIC ELECTIVE (DSE) COURSE-I
(Select Any One of the Following)

GEOG 106 (DSE I)
GEOGRAPHY AND ECOSYSTEMS

Course Code: GEOG 106
Course Type: DSE-I
Course Credit: 4
Teaching Hours/Week: 4

Evaluation: Maximum Marks: 100
Terminal Examination: 80 Marks
Internal Assessment: 20 Marks
Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Comprehend key environmental concepts, processes, interactions and linkages
- Analyse human-environment associations, ecological stability and disturbance
- Evaluate modifications in biomes and links to human induced changes
- Critique policies and devise initiatives for restoring ecological sustainability

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 short questions in about 25-30 words each. Each question shall carry 2 marks (Total 20 marks).
2. A total of 8 questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 15 marks (Total 60 marks). These will be in addition to compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. **The paper setter must put note (4) in the question paper.**

COURSE CONTENT

Unit –I

Foundations to Ecosystem and Geography

i. Ecosystem Approach: Definition, Characteristics and Components.
ii. Functional Characteristics: Food chains/webs; Trophic Levels and Ecological Pyramids
iii. Processes: Energy Flow, Nutrient Cycling, Ecological Productivity, Ecological Succession

Unit II

Contemporary Themes and Concepts:

iv. Ecosystem Stability and Disturbance; Ecological Change over Time and Space
v. Biodiversity: Importance; Gradients, Causes of Biodiversity Loss and Hot spots.
vi. Ecological and Carbon Footprints; Environmental Sustainability

Unit –III

Understanding the Nature of Biomes

vii. Forest Ecosystem: Tropical, Temperate and Boreal
viii. Grassland Ecosystem: Tropical Grassland and Temperate Grassland
ix. Desert Ecosystem: Hot Desert and Cold Desert.

Unit –IV

Case Studies: Human induced challenges and Issues

x. Hill Ecosystems with specific reference to the Siwaliks
xi. Wetland Ecosystems with specific reference Punjab Wetlands
xii. Agricultural Ecosystems with specific reference to the Green Revolution in Punjab.

LIST OF READINGS

Essential Readings:

1. Allison, S.K. and Stephen D. Murphy: *Routledge Handbook of Ecological and Environmental Restoration*, Routledge, 2017.
2. Brar, K. K.: *Green Revolution: Ecological Implications*, Dominant Publishers, Delhi, 1999.
3. MacDonald, Glen: *Biogeography: Introduction to Space, Time, and Life*, John Wiley, New York, 2003.
4. Odum, E.P. and Barrett, G.W.: *Fundamentals of Ecology*, 5th edition, Cengage Learning India Private Limited: New Delhi, 2009.
5. Singh, Savindra: *Environmental Geography*, Pravalika Publications, Allahabad, 2018.

Suggested Readings:

1. Ellis, Erle C., Kees Klein Goldewijk, Stefan Siebert, Deborah Lightman and Navin Ramankutty: Anthropogenic transformation of the biomes, 1700 to 2000. *Global Ecology and Biogeography.*19: 589–606, 2010.
2. Mathur, H.S.: *Essentials of Biogeography,* Pointer Publishers: Jaipur, 1988.
3. Robinson, H.: *Biogeography,* The English Language Book Society and Macdonald and Evans: London, 1982.
4. Steffen, W.: Jacques Grinevals, Paul Crutzen and J. McNeil. The Anthropocene: Conceptual and Historical Perspectives. *Philosophical Transactions of the Royal Society, A*. 369: 842-867. 2011
5. Tivy, Joy: *Biogeography:A Study of Plants in the Ecosphere,* Longman Scientific & Technical, U.K., 1993.

OR

GEOG 106 (DSE I)
POPULATION GEOGRAPHY

Course Code: GEOG 106
Course Type: DSE-I
Course Credit: 4
Teaching Hours/Week: 4

Evaluation: Maximum Marks: 100
Terminal Examination: 80 Marks
Internal Assessment: 20 Marks
Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the semester, students will be able to:

- Examine geographical approach to the study of population.
- Distinguish the concept and measurement of select population attributes
- Analyse the spatial patterns of population attributes.
- Evaluate population resource relationship.

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 short questions in about 25-30 words each. Each question shall carry 2 marks (Total 20 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 15 marks (Total 60 marks). These will be in addition to compulsory question at serial number 1.

3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. **The paper setter must put note (4) in the question paper.**

COURSE CONTENT

Unit -I

i. Nature, scope and approaches to population geography; population geography in India.
ii. Sources, quality and reliability of population data: census; surveys (national sample survey, national family health survey etc.); registration (national population register, vital statistics registration) miscellaneous sources.

Unit -II

iii. Global Demographic Scenario
iv. Concepts, determinants and world patterns of following attributes of population:
    - Distribution, Density, and Growth
    - Fertility, Mortality and Migration (including laws of Migration)
    - Age and Sex Composition

Unit -III

v. Linkages between Population Growth and Economic Development.
vi. Population Growth and Resource Scarcity (supply induced, demand induced; and structural scarcity) and Food Security

Unit -IV

vii. Population problems and policies of developed and less developed countries
viii. Population problems and policies of Canada and India.

LIST OF READINGS

Essential Readings:

1. Barret, H.R.: Population Geography, Oxford & Boyd, Oxford, 1997.
2. Beaujeu Garnier, J.: Geography of Population, Longman, London, 1966.
3. Chandna, R.C.: Geography of Population, Part I (13th Ed.) Kalyani Publishers, New Delhi, 2021.
4. Chandna, R.C.: Geography of Population, Part II (13th Ed.) Kalyani Publishers, New Delhi, 2021.
5. Chandna, R.C.: Geography of Population, Part I (Hindi, 12th Edition) Kalyani Publishers, New Delhi, 2022.
6. Chandna, R.C.: Geography of Population, Part II (Hindi, 12th Edition) Kalyani Publishers, New Delhi, 2022.
7. Demko, G.J. et. al: Population Geography: A Reader, McGraw Hill Books Co., New York, 2002.
8. Ghosh, B.N.: Fundamentals of Population Geography, Sterling Publication, 2005.
9. Hussain, M.: Population Geography, Rawat Publication, 2005.
10. Jones, H.R.: A Population Geography, Harper and Row Publishers, London, 1990.
11. Maurya, S.D.: Population Geography, Pravalika Publication, Allahabad, 2018
12. Newbold, K. Bruce, Population Geography, Rawat Publications, 2017.
13. Quazi, S.A.: Population Geography, APH Publishing Corporation, 2012.
14. Trewartha, G.T.: The More Developed Realm: A Geography of its Population, Pergamon Press, New York, 1978.
15. Weeks, R. John: Population: An Introduction to Concepts and Issues, Wadsworth Thomson Learning, Toronto, Canada, 11th Edition, 2012.

Suggested Readings:

1. Boyle, P.: Population Geography: Migration and Inequalities in Mortality and Morbidity, Progress in Human Geography, Sage Publications, 2004, http:// phg.sagepub.com
2. Chandna, R.C.: Environmental Awareness, Kalyani Publishers, New Delhi, 1998.

3. Crook, Nigel: Principles of Population and Development, edited by Ian, M. Timacus, Oxford University Press, Oxford, 1997.
4. Hugo, G.: Population Geography Progress in Human Geography, Sage Publications, 2007.
5. Peters, L. Gary: Population Geography: Problem, Concepts and Prospects, Kendeall/Hent Publishing Company, 2008.
6. Premi, M.K.: India's Changing Population Profile, National Book Trust India, New Delhi, 2009.
7. United Nations: World Population Situation in 1983, U.N. Publications, New York, 1984.
8. Woods, R.: Population Analysis in Geography, Longman, London, 1979.

DISCIPLINE SPECIFIC ELECTIVE (DSE) COURSE-II
(Select Any One of the Following)

GEOG 107 (DSE II)
CULTURAL GEOGRAPHY

Course Code: GEOG 107 ***Evaluation: Maximum Marks: 100***

Course Type: DSE-II — Terminal Examination: 80 Marks
Course Credit: 4 — Internal Assessment: 20 Marks
Teaching Hours/Week: 4 — Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Explain the concepts, processes and manifestations associated withCultural Geography.
- Delineate the processes that have resulted in the production of a particular landscape.
- Critique the role of power in the construction of cultural landscapes.
- Develop a sensitivity towards cultural pluralism and social diversity.

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 short questions in about 25-30 words each. Each question shall carry 2 marks (Total 20 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 15 marks (Total 60 marks). These will be in addition to compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. **The paper setter must put note (4) in the question paper.**

COURSE CONTENT

Unit-I

i. CulturalGeography:Definition,EvolutionandDevelopment
ii. PlaceofCulturalGeographywithinGeography
iii. DoingCulturalGeography:PhilosophicalBasesandMethodologies

Unit-II

iv. CulturalHearth,CulturalRegion,Landscape,Environment
v. Nature/Human;Space/Place;Regionalism,Homeland
vi. Identity&Difference;Power&Discourse;Hegemony

Unit-III

vii. LandscapeEvolution;CulturalDiffusion;Adaptation
viii. Acculturation,Assimilation;Resistance
ix. Representation;Construction;Socio-spatialDialectic

Unit-IV

x. TransformingLandscapes
xi. OurUnequalWorld
xii. TheProductionofKnowledgeandRepresentationinCulturalGeography

LISTOFREADINGS

EssentialReadings:

1. Anderson, Jon: Understanding Cultural Geography Places and Traces, London: Routledge, 2010.
2. Anderson, K. Domosh, M., Pile, S. & Thrift, N. (eds.): Handbook of Cultural Geography, London: Sage Publications, 2003.
3. Bertolas, R. J. (1998): Cross-cultural environmental perception of wilderness. Professional Geographer, 50(1), pp 98-111.
4. Cosgrove, D. & Jackson, P. (1987): New directions in cultural geography. Area, 19(2): 95-101
5. Norton, W. and Walton-Roberts, M.: Cultural Geography: Environments, Landscapes, Identities, Inequalities. Ontario: Oxford University Press, 2014.
6. Price, M. & Lewis, M. (1993): The reinvention of cultural geography. Annals of the Association of American Geographers, 83 (1), pp1-17.
7. Shurmer-Smith, P (ed.). Doing Cultural Geography, Sage, New Delhi, 2003.

SuggestedReadings:

1. Oakesand Price(eds.): The Cultural Geography Reader, Routledge, London & New York, 2013.
2. Massey, D.: Space, Place and Gender. Cambridge: Polity Press, 1994
3. Mitchell, D.: Cultural Geography: A Critical Introduction. Malden: Blackwell, 2000
4. Sauer, C.: The Morphology of Landscape. Berkeley: University of California Press, 1925
5. Shields, R.: Places on the Margin: Alternative Geographies of Modernity. London and New York: Routledge, 1991

OR

GEOG 107 (DSE II)
ELECTORAL GEOGRAPHY

Course Code: GEOG 107
Course Type: DSE -II
Course Credit: 4
Teaching Hours/Week: 4

Evaluation: Maximum Marks: 100
Terminal Examination: 80 Marks
Internal Assessment: 20 Marks
Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Understand spatial dimensions of varied nature of constituencies.
- Delimit and demarcate constituencies for different purposes.
- Appreciate problems associated with voting, voters, contesting candidates and political parties.
- Suggest the electoral reforms in evolving scenario.

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 short questions in about 25-30 words each. Each question shall carry 2 marks (Total 20 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 15 marks (Total 60 marks). These will be in addition to compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. **The paper setter must put note (4) in the question paper.**

COURSE CONTENT

Unit I

i. Geography and Electoral Studies
ii. Electoral Systems and Electoral Data
iii. Constituencies, Delimitation methods

Unit II

iv. Geography of Voting: Political Cleavage and Geographies of Party Support
v. Social Bases of Geographies of Voting

Unit III

vi. Geographical Influences in Voting: Neighbourhood Effect, Candidates, Issues and Campaigns
vii. Geography of Representation
viii. Electoral Abuses, Electoral Reform

Unit IV

ix. Towards a Geography of Power: Geography, Elections, Representation and Power
x. Gerrymandering
xi. Methods of casting vote

LIST OF READINGS

Essential Readings

1. Harold Clarke, David Sanders, Performance Politics and the British Voter, Cambridge: Marianne Stewart & Paul Whiteley Cambridge University Press, 2009.
2. Russell Dalton Citizen Politics: Public Opinion and Political Parties in Advanced Industrial Democracies 5th ed., Washington: CQ Press, 2008.
3. David Denver Elections and Voters in Britain 2nd ed., Basingstoke: Palgrave Macmillan, 2007.
4. Denver, David. Electoral Geography and Electoral Systems in Elections and Voters in Britain 2nd ed., Basingstoke: Palgrave Macmillan, 2006. Chapter 7.
5. Evans, Jocelyn. Voting and the economy in Voters and Voting: An Introduction, London, Sage, 2004. Chapter 6.
6. Lewis Beck, Michael & Mary Stegmaier. American Voter to Economic Voter: Evolution of an Idea, Electoral Studies, 28, 2009, 625–631.

Suggested Readings

1. Aldrich, John. Rational Choice and Turnout. American Journal of Political Science, 37(1): 246-278. 1993.
2. Clarke, Harold, David Sanders. The American voter's British cousin, Electoral Studies, Marianne Stewart & Paul Whiteley 2009, 28(4): 632-641.
3. Dalton, Russell. Elections and Political Parties, in Citizen Politics 5th ed., Washington: CQ Press, 2008. Chapter 7.
4. Evans, Jocelyn. Introduction & the historical development of voting studies in Voters and Voting: An Introduction, London, Sage, 2004. Chapters 1 & 2.
5. Bartle, John & Samantha Laycock _Elections and Voting in Developments in British Politics edited by Patrick Dunleavy *et al*, Basingstoke: Palgrave Macmillan, 2006. Chapter 5.
6. Burnham, Peter, Kelly Gilland. Surveys and Opinion Polls in Research Methods in Wyn Grant & Zig Layton-Henry Politics 2nd ed., Basingstoke: Palgrave, 2008. Chapter 5.
7. Crewe, Ivor. The Opinion Polls: The Election They Got (Almost) Right, Parliamentary Affairs 58(4): 684-698, 2005.
8. Lewis-Beck, Michael. Election Forecasting: Principles and Practice, British Journal of Politics and International Relations, 7:2, 145-64. 2005.
9. Scarbrough, Elinor. The British Election Study and Electoral Research, Political Studies 48 (3): 391–414. 2000.
10. Sides, John. Electoral Politics in Peele, Bailey, Cain & Peters, eds., Developments in American Politics 5, Basingstoke: Palgrave Macmillan, 2006.
11. Whiteley, Paul _Forecasting Seats from Votes in British General Elections, British Journal of Politics and International Relations, 7:2, 165-73. 2005.

Semester-II

DISCIPLINE SPECIFIC CORE (DSC) COURSES

GEOG 201
GEOGRAPHY OF INDIA

Course Code: GEOG 201 ***Evaluation: Maximum Marks: 100***

Course Type: DSC — Terminal Examination: 80 Marks

Course Credit: 4 — Internal Assessment: 20 Marks

Teaching Hours/Week: 4 — Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Appreciate the physiographic, social and economic diversities in India
- Apply regionalisation techniques in delineating India's physical and socio-cultural regions.
- Develop the idea of national unity by synthesising various regional dimensions.

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 short questions in about 25-30 words each. Each question shall carry 2 marks (Total 20 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 15 marks (Total 60 marks). These will be in addition to compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. **The paper setter must put note (4) in the question paper.**

COURSE CONTENT

Unit-I

i. Unity in diversity of India: Unifying mechanism and divisive streaks.
ii. Evolution of the administrative map of India since Independence.

Unit-II

iii. Role of language, religion and culture in the formation of regions.
iv. The question of regional disparity and identity in India.

Unit-III

v. Regionalisation schemes of India: Physiographic (SP Chatterjee), Climatic (Koeppen and Trewartha), Agricultural (Jasbir Singh and CB. Mamoria), and Industrial (BN Sinha).

Unit-IV

vi. Northwest India as a Geographic Entity: Jammu & Kashmir, Ladakh, Himachal Pradesh, Haryana, Punjab and Union Territories of Delhi and Chandigarh.
vii. Land: Physiography and drainage.
viii. People: Population number, distribution and density, growth and urbanization.
ix. Economy: Agriculture, Industry and Transport.

LIST OF READINGS

Essential Readings:

1. Ahmad Aijazuddin: *Social Geography*, Rawat Publication, New Delhi, 1999.
2. Chandna, R.C.: *Geography of Population*, Kalyani Publishers, Delhi, 1998.
3. Gautam, A.: *Advanced Geography of India*, Sharda Pustak Bhawan, Allahabad, 2009.
4. Hussain, M.: *Geography of India*, Tata McGraw Hill Pub. Company Ltd., New Delhi, 2008.

5. Krishan, G.: *The Vitality of India: A Regional Perspective*, Rawat Publications, 2017
6. Singh, J.: *India-A Comprehensive Systematic Geography*, Gyanodya Prakashan, Gorakhpur, 2003.
7. Spate O.H.K. & A.T.A. Learmonth: *Geography of India and Pakistan*, Methuen, London (First Indian Edition,1984, Munshiram Manoharlal, New Delhi), 1967.
8. Sukhwal, B. L.: *India: A Political Geography*, Allied Publishers, New Delhi.
9. Tirtha, Ranjit: *Emerging India*, Conpub. Ann Arbour, U.S.A. Michigan, 2006.
10. Tiwari, R.C.: *Geography of India*, Prayag Publishers, Allahabad, 1999.

Suggested Readings:

1. Center for Science and Environment: *State of India's Environment*, New Delhi, 1985.
2. Desai, B. Sonalde et al: *Human Development in India*, Oxford Univ. Press, New Delhi, 2010.
3. Dreze, Jean and Amartya Sen: *Indian Development*, Oxford University Press, Delhi, 1996.
4. Farmer, B. H.: *Introduction to South Asia*, Methuen, London, 1983.
5. Francis, Robinson: *The Cambridge Encyclopedia of India, Pakistan, Bangladesh, Sri Lanka, Nepal, Bhutan, and the Maldives*, Cambridge University Press, London, 1989.
6. Gazetteer of India, Vol. I & II: *Ministry of Information & Broadcasting*, GOI, New Delhi, 1973.
7. Johnson, B.L.C: *India: Resources and Development*, Arnold Heinemann, London, 1980.
8. Sivaramkrishnan, K.C., Amitabh Kundu and B.N. Singh: *Oxford Handbook of Urbanisation in India,* Oxford University Press, New Delhi, 2007.

GEOG 202
CLIMATOLOGY

Course Code: GEOG 202 | ***Evaluation: Maximum Marks: 50***

Course Type: DSC | Terminal Examination: 40 Marks

Course Credit: 2 | Internal Assessment: 10 Marks

Teaching Hours/Week: 2 | Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the semester, the students will be able to:

- ❖ Comprehend atmospheric phenomena and dynamics.
- ❖ Evaluate the interactions between atmospheric processes and weather phenomena
- ❖ Synthesize climate and weather dynamics and classify climatic regions

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 6 short questions in about 25-30 words each. Each question shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. **The paper setter must put note (4) in the question paper.**

COURSE CONTENT

Unit -I

i. The Earth's Atmosphere: Nature, Composition and Structure.
ii. Solar Radiation: EMR, Electromagnetic Spectrum, Radiation Laws
iii. Factors influencing Insolation, Mechanisms of Heat Transfer; Global Distribution; Heat Budget

Unit -II

i. Temperature: Concept, Measurement, Scales; Vertical and Horizontal Distribution.
ii. AtmosphericHumidity: Sources, Measurement, and Influencing factors
iii. Precipitation: Measurements, Forms,and Global Distribution.

Unit-III

i. Atmospheric Pressure: Influencing Factors, Global Pressure belts and circulation
ii. Jet Streams and Polar Vortex: Formation, Characteristics and Impacts on weather
iii. Tropical and Mid-Latitude Cyclones: Formation, Growth Mechanisms and Effects

Unit-IV

iv. Indian Monsson: Concept, Features, Types, and Mechanism
v. Drivers of Indian Monsoon; MJO, ENSO and IOD Linkages
vi. Western Disturbances in India: Origin, Features and Impacts

LIST OF READINGS

Essential Readings:

1. Barry, R.G. & Chorley, R.J.: Atmosphere, Weather and Climate, 9th Edition, Methuen Co. Ltd., London, 2009.
2. Bhutani, Smita: Our Atmosphere, Kalyani Publishers, Ludhiana, 2000.
3. Frederick K. Lutgens & Edward J. Tarbuck: The Atmosphere: An Introduction to Meteorology, Prentice Hall of India Pvt. Ltd., New Delhi, 2018.
4. Lal, D.S., Climatology, Sharda Pustak Bhaban, Allahabad, 2009.
5. Strahler, A. and A. Strahler: Introducing Physical Geography, 6th Edition, John Wiley & Sons, Hoboken, New Jersey, 2013.

Suggested Readings

1. Thompson R.D.: Applied Climatology - Principles & Practice, John Wiley, New York, 1997.
2. Ahrens, C. D. and Henson, R.: Meteorology Today: An Introduction to Weather, Climate and the Environment, 12th Edition, Cengage Learning, USA, 2018.
3. Schneider, S. H., Root, T. L., and Mastrandrea, M. D. (eds.): Encyclopedia of Climate and Weather, 2nd Edition, Oxford University Press, New York, 2011.
4. Robinson, P. J. and Henderson- Sellers, A.: Contemporary Climatology, 2nd edition, Pearson Education Ltd., Harlow, UK, 1999.
5. Rohli, R. V. and Vega, A. J.: Climatology, 3rd Ed., Jones and Bartlett Learnings, Burlington, 2015.

GEOG 203 (Practical)
CLIMATE AND WEATHER ANALYSIS

Course Code: GEOG 203 (Practical)
Course Type: DSC
Course Credit: 2
Teaching Hours/Week: 4

Evaluation: Maximum Marks: 50
Practical Record: 30 Marks
Viva-Voce: 10 Marks
Internal Assessment: 10 Marks
Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Inspect, Categorise and Describe Climate and Weather Parameters
- Analyse and Measure Weather and Climate Patterns, Trends, and Extremes
- Interpret and Forecast Using Weather Reports and Doppler Radar Imageries
- Understand and Interpret Weather Maps

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty member teaching the course at least 10 days before the commencement of end-semester examination.
2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examinations.
3. Internal assessment may include written assignments/term paper, participation in class discussion, and attendance etc.

COURSE CONTENT: Practical Exercises/Lab Work

1. Representation of Temperature and Rainfall Data: Line & Bar Graph, Hyetograph
2. Climograph and Hythergraph: Construction and Interpretation
3. Drawing Isohyets and Isotherms: Interpolation Techniques
4. Statistical Analysis of Climate Data: Trend and Variability
5. Measurement ofClimate Extreme Indices: Temperature and Precipitation
6. Reading digital weather maps/apps/Doppler Radar Imagery

LIST OF READINGS

Essential Readings:

1. Barry, R.G. & Chorley, R.J.: Atmosphere, Weather and Climate, 9th Edition, Methuen Co. Ltd., London, 2009.
2. Frederick K. Lutgens & Edward J. Tarbuck: The Atmosphere: An Introduction to Meteorology, Prentice Hall of India Pvt. Ltd., New Delhi, 2012.
3. Strahler, A. and A. Strahler: Introducing Physical Geography, 6th Edition, John Wiley & Sons, Hoboken, New Jersey, 2013.

Suggested Readings

1. Ahrens, C. D. and Henson, R.: Meteorology Today: An Introduction to Weather, Climate and the Environment, 12th Edition, Cengage Learning, USA, 2018.
2. Schneider, S. H., Root, T. L., and Mastrandrea, M. D. (eds.): Encyclopedia of Climate and Weather, 2nd Edition, Oxford University Press, New York, 2011.
3. Rohli and Vega: Climatology, 3rd Ed., Jones and Bartlett Learnings, Burlington, 2015.

GEOG 204
FUNDAMENTALS OF REMOTE SENSING

Course Code: GEOG 204
Course Type: DSC
Course Credit: 2
Teaching Hours/Week: 2

Evaluation: Maximum Marks: 50
Terminal Examination: 40 Marks
Internal Assessment: 10 Marks
Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- ❖ Recount the principles of remote sensing and its applications
- ❖ Analyse and interpret information from remotely sensed data
- ❖ Synthesize remotely sensed information for its applications/usages

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 6 short questions in about 25-30 words each. Each question shall carry 2 marks (Total 12 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 7 marks (Total 28 marks). These will be in addition to compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. **The paper setter must put note (4) in the question paper.**

COURSE CONTENT

Unit-I

i. Remote Sensing: Concept, Types and Applications
ii. Remote Sensing Platforms, Sensors and Scanning Systems
iii. Introduction to Major Remote Sensing Satellites

Unit-II

iv. Radiation Principles; Electromagnetic Spectrum
v. Energy-Atmosphere Interaction; Atmospheric Windows
vi. Energy-Earth Interaction; Spectral Signatures of Surface Features

Unit-III

vii. Image: Meaning and Types and Characteristics
viii. Resolution: Spatial, Spectral, Radiometric and Temporal
ix. Basics of Image Processing; Elements of Image Interpretation.

Unit-IV

x. Thermal Remote Sensing: Concept, Mechanism and Application
xi. Microwave Remote Sensing: Concept, Mechanism and Application
xii. Hyperspectral Remote Sensing: Concept, Mechanism and Application

LIST OF READINGS

Essential Readings:

1. Campbell, J. B. and R.H. Wynne: *Introduction to Remote Sensing,* 5th Ed., Guilford Press, 2012.
2. Gomarasca, Mario A, *Basics of Geomatics*, Springer: Heidelberg, 2009.
3. Joseph, G.: *Fundamentals of Remote Sensing*, Universities Press, 2005.
4. Rees, W.G.: *Physical Principles of Remote Sensing,* 3rd Ed., Cambridge University Press, 2012.
5. Schowengerdt, Robert A.: *Remote Sensing: Models and Methods for Image Processing*, Second Edition, Academic Press: San Diego, 2017.
6. Srivastava, G.S.: *An Introduction to Geoinformatics*, McGraw Hill Edu., India, New Delhi, 2014.
7. Dong and Qi Chen: *LiDAR Remote Sensing and Applications*, CRC Press, Taylor & Francis, 2018

Suggested Readings:

1. Jensen, J.R.: *Introductory Digital Image Processing: A Remote Sensing Perspective*, Pearson Prentice Hall, 2005.
2. Lillesand, T.M., R.W. Kiefer and J.W. Chipman: *Remote Sensing and Image Interpretation*, 5th Edition, John Wiley and Sons, 2004.
3. Mather, P.M. and M.G. Koch: *Computer Processing of Remotely-Sensed Images: An Introduction,* John Wiley & Sons, 2011.
4. Nag, P. and M. Kudrat: *Digital Remote Sensing*, Concept Publishing Company, New Delhi, 1998.
5. Njoku, E.G.: *Encyclopaedia of Remote Sensing*, Springer, New York, 2014.
6. Richards, John A.: *Remote Sensing Digital Image Analysis: An Introduction,* 5th Edition, Springer-Verlag Berlin Heidelberg, 2013.

GEOG 205 (Practical)
DIGITAL IMAGE PROCESSING

Course Code: GEOG 205 (Practical) ***Evaluation: Maximum Marks: 50***

Course Type: DSC — Practical Record: 30 Marks
Course Credit: 2 — Viva-Voce: 10 Marks
Teaching Hours/Week: 4 — Internal Assessment: 10 Marks
Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Comprehend satellite data analysis through focused hands-on-training.
- Perform technical procedures of data analysis and extraction
- Compare and evaluate information from remotely sensed imageries
- Synthesize image interpretation for various applications

INSTRUCTIONS FOR EXAMINERS:

1. Each candidate shall prepare a Practical File under the supervision and guidance of class teacher. The candidates shall submit Practical File duly approved and signed by the faculty member teaching the course at least 10 days before the commencement of end-semester examination.

2. Assessment of practical record and viva voce will be done by a Board of Examiners, consisting of one external examiner and one internal examiner, at the time of practical examination.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

COURSE CONTENT: List of Practical/Lab Exercises:

1. Measuring Satellite Orbital Characteristics
2. EMR: Wavelength, Frequency and Energy Characteristics
3. Spectral Signature of Surface Features
4. Visual Interpretation of Satellite Imageries
5. Image Processing:Polynomial Rectification-Georeferencing
6. Preparing Image Subset and Mosaic
7. Digital Image Interpretation; Pixel and Histogram Analysis
8. Supervised Classification and Accuracy Assessment
9. Unsupervised Classification and Accuracy Assessment
10. Satellite Data based Indices: NDVI, NDMI, EVI, NDWI, NDBI, VCI, TCI.

LIST OF READINGS

Essential Readings:

1. Campbell, J. B. and R.H. Wynne: *Introduction to Remote Sensing,* 5th Ed., Guilford Press, 2012.
2. Gomarasca, Mario A, *Basics of Geomatics*, Springer: Heidelberg, 2009.
3. Joseph, G.: *Fundamentals of Remote Sensing*, Universities Press, 2005.
4. Schowengerdt, Robert A.: *Remote Sensing: Models and Methods for Image Processing*, Second Edition, Academic Press: San Diego, 2017.
5. Srivastava, G.S.: *An Introduction to Geoinformatics*, McGraw Hill Edu., India, New Delhi, 2014.

Suggested Readings:

1. Jensen, J.R.: *Introductory Digital Image Processing: A Remote Sensing Perspective*, Pearson Prentice Hall, 2005.
2. Lillesand, T.M., R.W. Kiefer and J.W. Chipman: *Remote Sensing and Image Interpretation*, 5th Edition, John Wiley and Sons, 2004.
3. Mather, P.M. and M.G. Koch: *Computer Processing of Remotely-Sensed Images: An Introduction,* John Wiley & Sons, 2011.
4. Njoku, E.G.: *Encyclopaedia of Remote Sensing*, Springer, New York, 2014.
5. Richards, John A.: *Remote Sensing Digital Image Analysis: An Introduction,* 5th Edition, Springer-Verlag Berlin Heidelberg, 2013.

DISCIPLINE SPECIFIC ELECTIVE (DSE) COURSE-I
(Select Any One of the Following)

GEOG 206 (DSE-I)
AGRICULTURE GEOGRAPHY

Course Code: GEOG 206
Course Type: DSE-I
Course Credit: 4
Teaching Hours/Week: 4

Evaluation: Maximum Marks: 100
Terminal Examination: 80 Marks
Internal Assessment: 20 Marks
Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, students will be able to:

- Comprehend and analyse basic concepts of agricultural geography.
- Critique the principles of agricultural, land use and land capability classifications.
- Regionalize and classify agricultural patterns in real time.
- Apply the knowledge gained to decode regional and national agriculture scenario.

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 short questions in about 25-30 words each. Each question shall carry 2 marks (Total 20 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 15 marks (Total 60 marks). These will be in addition to compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. **The paper setter must put note (4) in the question paper.**

COURSE CONTENT

Unit-I

i. The nature, subject matter and progress in Agricultural Geography
ii. Approaches:(a) commodity, (b) systematic, (c) regional
iii. Determinants:(a) physical, (b) economic, (c) socio-cultural

Unit-II

iv. Selected agricultural concepts and their measurement
   a. Intensity of cropping
   b. Degree of commercialization
   c. Diversification and specialization
   d. Efficiency and productivity.

Unit-III

v. Land-use survey and classification (British and Indian)
vi. Land capability classification (U.S. and Britain)

Unit-IV

vii. Critical evaluation: Classification of the World Agriculture w.r.t. to Whittlesey
viii. New perspectives: Contract Farming, Agri-business and Food Security

LIST OF READINGS

Essential Readings

1. Husain, M.: *Systematic Agricultural Geography*, Rawat Publications, Jaipur,2004.
2. Ilbery, B.W.: *Agricultural Geography,* Oxford University Press, Oxford,1985.
3. Pacione, Michael: *Progress in Agricultural Geography*, Routledge Revivals,2013.
4. Singh, J .and Dhillon, S. S.: *Agricultural Geography*, Tata McGraw Hill, New Delhi, 1984.
5. Singh, Jasbir: *AgriculturalGeography,* 3rd edition, Oxford, New Delhi, 2003.

Suggested Readings:

1. Bryant, Adler: *Agricultural Economics and Agribusiness Management,* Syrawood Publishing House, 2016.
2. Gosal, G.S. and Krishan, Gopal: *Regional Disparities in Levels of Socio-Economic Development in Punjab,* Vishal Publications, Kurukshetra,1984.
3. Grigg, D.B.: *The Agricultural Systems of the World: An Evolutionary Approach,* Cambridge University Press, Cambridge, 1978.
4. Pacione, Michael: Progress in Agricultural Geography, Routledge, 2014.
5. Shafi, M.: *Agricultural Productivity and Regional Imbalances*, Concept, New Delhi, 1984.
6. Singh, Jasbir: *Dynamics of Agricultural Change*, Oxford, NewDelhi,1990.
7. Whealler, K.E., Ladley, A.M. and Leong, F.C.: *Studies in Agricultural Geography,* Bland Educational, London, 1970.

OR

GEOG 206 (DSE-I)
HAZARDS AND DISASTERS

Course Code: GEOG 206 ***Evaluation: Maximum Marks: 100***
Course Type: DSE-I Terminal Examination: 80 Marks
Course Credit: 4 Internal Assessment: 20 Marks
Teaching Hours/Week: 4 Examination Time: 3 Hours

COURSE OUTCOMES: At the end of this course, the students will be able to:

- Evaluate hazard and disaster mechanisms to determine disaster risk
- Determine disaster resilience of natural and human systems.
- Design mechanisms to ensure a safe and resilient society
- Critical evaluation of disaster management plans and policies

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 short questions in about 25-30 words each. Each question shall carry 2 marks (Total 20 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 15 marks (Total 60 marks). These will be in addition to compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. **The paper setter must put note (4) in the question paper.**

COURSE CONTENT

Unit-I

i. Basic Concepts: Hazard, Vulnerability, Risk, Disaster
ii. Classification/Types of Hazards/Disasters
iii. Evolution of Disaster Studies and its Current Status

Unit-II

iv. Earthquake and Landslide
v. Floods, Cyclone, and Drought
vi. Regional Dimension of Hazards/Disasters in India.

Unit-III

vii. Disaster Risk Reduction (DRR), Global Initiatives for Risk Reduction
viii. Community Based Disaster Risk Management (CBDRM)
ix. Gender-Sensitive Disaster Risk Reduction

Unit-IV

x. Disaster Management: Pre-disaster; Emergency and Post-disaster phase
xi. Disaster Management Mechanism in India
xii. Disaster Management Plans and Policies in India

LIST OF READINGS

Essential Readings:

1. Blaikie, P. and Others (1994). *At Risk: Natural Hazards, People's Vulnerability, and Disasters,* Routledge, London.
2. Birkmann, J. (2006). *Measuring Vulnerability to Natural Hazards: Towards Disaster Resilient Societies*. US: United Nations University Press.
3. Coppola, Damon P. (2015). *Introduction to International Disaster Management*. Third Edition, Elsevier, London.
4. Gupta, M.C.; L.C. Gupta, B.K. Tamini, Vinod K. Sharma (2000). *Manual on Natural Disaster Management in India,* National *Disaster Management Centre, New Delhi*

5. Hyndman, D. and D. Hyndman. (2009). *Natural Hazards and Disasters*. 2nd edition. USA, Belmont: Brooks/Cole.
6. Madu, Christian N and Chu-hua Kuei (2018). *Handbook of Disaster Risk Reduction and Management*. World Scientific Publishing Co. Pte. Ltd, Singapore.

Suggested Readings

1. Alexander, D. E. (1993). *Natural Disasters*. London: University College London Press and; Dordrecht and Boston: Kluwer Academic Publishers.
2. Alexander, D. E. (2000). *Confronting Catastrophe: New Perspectives on Natural Disasters*. Harpenden, U.K: Terra Publishing.
3. Burton, I., Kates, R.W. and White, G.F. (1993). *Environment as Hazard*, 2nd edition, Guilford Press, New York.
4. Hewitt, K. (1997). *Regions of Risk: A Geographical Introduction to Disasters*, Longman, London.
5. Kasperson, J.X., Kasperson, R.E. and Turner, B. L. (1995). *Regions at Risk: Comparisons of Threatened Environments, United Nation* University Press, Tokyo.
6. Schneider, S.K. (1995). *Flirting with Disaster: Public Management in Crisis Situations*, M.E. Sharpe, New York

DISCIPLINE SPECIFIC ELECTIVE (DSE) COURSE-II
(Select Any One of the Following)

GEOG 207 (DSE-II)
ADMINISTRATIVE GEOGRAPHY

Course Code: GEOG 207
Course Type: DSE-II
Course Credit: 4
Teaching Hours/Week: 4

Evaluation: Maximum Marks: 100
Terminal Examination: 80 Marks
Internal Assessment: 20 Marks
Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Appreciate the role of Geography in area administration
- Apply techniques for analysing the structural attributes of administrative areas
- Analyse administrative hierarchy, size, shape and location of headquarters
- Develop insights to various public policies in area administration

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 short questions in about 25-30 words each. Each question shall carry 2 marks (Total 20 marks).
2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 15 marks (Total 60 marks). These will be in addition to compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. **The paper setter must put note (4) in the question paper.**

COURSE CONTENT

Unit-I

i. Administrative Geography; Definition, subject matter and significance: Interface between Geography and Public Administration and Political Geography.

Unit-II

ii. Study of administrative areas in terms of (a) Evolution, (b) Nature, (c) Structural Attributes (hierarchy, size, shape and headquarters) and (d) Administrative area reform.

Unit-III

iii. Area Administration: (a) Public Policy: Formulation, Implementation and Impact, (b) Public Finance: Public goods and public economy.

Unit-IV

iv. Administrative System: (a) The world pattern, (b) Case studies: India and U.S.A.

LIST OF READINGS

Essential Readings:

1. Bennett, R.J.: *Geography of Public Finance*, Methuen, New York, 1980.
2. Dorling, D., & Shaw, M.: *Geographies of the agenda: public policy, the discipline and its (re) 'turns'*, Progress in Human Geography, 26(5), pp 629-641, 2002.
3. Kant, Surya: *Administrative Geography of India*, Rawat Publication, Jaipur, 1988.
4. Krishan, Gopal: *Administrative Geography*, Transaction of the Institute of Indian Geographers, Vol. 5, No. 2, pp 101-108, 1983.
5. Krishan, Gopal: *The World Pattern of Administrative Area Reform*, The Geographical Journal, Vol. 154, No. 1, pp 93-99, 1988.
6. Martin, R.: *Geography and public policy: the case of the missing agenda,* Progress in Human Geography, 25(2), pp 189-210, 2001.
7. Singh, M. P.*: Reorganization of States in India,* Economic and Political Weekly, pp 70-75, March 15, 2008.
8. Surya Kant: *Organisation of space for administration: Presidential Address*, Transaction Institute of Indian Geographers, Vol.39 (2), pp 163-169, July 2017.
9. Surya Kant and Nina Singh (*eds)*: *Geography, Development, Public Policy: Select Essays of Gopal Krishan,* R.K. Books, New Delhi, 2017.
10. Ward, K.: *Geography and public policy: A Recent history of Policy Relevance.* Progress in Human Geography, 29(3), pp 310-319, 2005.

Suggested Readings:

1. Alderfer, H.F.: *Local Government in Developing Countries*, McGraw Hill, New York, 1964.
2. Massam, B. H.: *The Spatial Structure of Administrative Systems*, Association of AmericanGeographers, Research Paper No. 12, 1972.
3. Muir, R.: *Modern Political Geography*, MacMillian, London, 1981.
4. Prescott, J.R.V.: *The Geography of the State Policies*, Hutchinson, London, 1968.

OR

GEOG 207
URBAN GEOGRAPHY (DSE-II)

Course Code: GEOG 207
Course Type: DSE-II
Course Credit: 4
Teaching Hours/Week: 4

Evaluation: Maximum Marks: 100
Terminal Examination: 80 Marks
Internal Assessment: 20 Marks
Examination Time: 3 Hours

COURSE OUTCOMES: At the end of the course, the students will be able to:

- Trace the evolution of and critique the theoretical foundations of Urban Geography.
- Compare and classify urban centres and urban systems across socio-cultural contexts.
- Critique the city as a lived space in the light of issues dealing with survival in and of cities.

INSTRUCTIONS FOR EXAMINERS:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The student shall attempt any 10 short questions in about 25-30 words each. Each question shall carry 2 marks (Total 20 marks).

2. A total of eight questions will be set out of the whole syllabus at least *two* from each unit. The candidates will attempt *four* questions selecting one from each unit and each question will carry 15 marks (Total 60 marks). These will be in addition to compulsory question at serial number 1.
3. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
4. For reappear/improvement candidates(s), who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be increased proportionally to maximum marks of the paper in lieu of internal assessment. **The paper setter must put note (4) in the question paper.**

COURSE CONTENT

Unit-I

i. UrbanGeography:AnIntroduction

- Definition,ScopeandApproaches.
- DevelopmentandRecentTrends.
- ContributionofIndianscholars.

Unit-II

ii. City: Attributes and Processes

- Setting: Location, Situation and Site (Definition, Types and Significance)
- Internal Structure: Theories of Internal Structure, Internal Structure of Indian Cities, Comparison of Internal Structure of Planned and Evolved Cities, Urban Fringe
- Classification of Urban Places: Non-Functional Classification, Functional Classification

Unit-III

iii. UrbanSystems

- UrbanSystems:Definition,ModelsofCity-SizeDistribution,UrbanSysteminIndia
- City-RegionRelations:BasisandNature,Definition,Demarcation,FunctionalStructure of Umland.
- CitiesasCentralPlaces:ModelsofChristallerandLosch

Unit-IV

iv. CityasLivedSpace

- NegotiatingtheCity
- ImaginingtheCity
- ContemporaryUrbanIssues:Unequal City,SustainabilityandtheCity,Governance,Violence in the City

LISTOFREADINGS

EssentialReadings:

1. Carter,Harold:*TheStudyofUrbanGeography*,EdwardArnold,London,4th Edition,1995.
2. Kaplan,D.,J.WheelerandS.Holloway.:*UrbanGeography*,Wiley,USA,2009.
3. Hubbard,P.:*City*,Routledge,LondonandNewYork,2006.
4. Ganeshwari,S.KahlonandV.Chandel.,TheCityAlive:HumanisingUrbanGeographicResearchinIndia,*Population Geography*,Vol.37(1&2),2015.
5. Hall,T.:*UrbanGeography*,Routledge,LondonandNewYork,2006.
6. Mayer,H.M.&C.F.Kohn:*ReadingsinUrbanGeography*,University Press,Chicago,1959.
7. Pacione,Michael:*UrbanGeography:AGlobalPerspective*,Routledge,NewYork,3rd Edition,2007.
8. Sharma, K.D.:*UrbanDevelopmentintheMetropolitanShadow*,Inter-India,NewDelhi, 1985.

FurtherReadings:

1. Paddison,R.:*HandbookofUrbanStudies*,Sage:NewDelhi,2001.
2. Westwood,S.andJ.Williams.:*ImaginingCities*,Routledge,LondonandNewYork,1997.
3. Kundu,A.:*UrbanDevelopmentandUrbanResearchinIndia*,KhannaPub.,NewDelhi,1992.
4. Short,JohnR.:*AnIntroductiontoUrbanGeography*,Routledge,London,1984.